

2.0 APPROACH IN PROBLEM SOLVING

COMPUTER SCIENCE 2

SC025 | SESI 2026/2027

NAME:

CLASS:

TOPIC 2: APPROACH IN PROBLEM SOLVING

1. Describe five (5) steps in problem solving.
   (a) Problem analysis

   (b) Design a Solution

   (c) Implementation

   (d) Testing

   (e) Documentation

2. Identify the step of problem solving based on the statement below.
   a. Process of translating the algorithm into the syntax of a given programming language.

   b. All of the flowcharts, messages, algorithms, lines, of code and the user manual are compiled together.

3. Identify input, process, and output for the following problem statements:
   a. Calculate and print the volume of a sphere.

   b. Reads three numbers and print their sum and average.

   c. Calculate the Body Mass Index (BMI) for a person.

   d. Calculate and display the overtime pay received by an employee. Overtime rate is RM5 per hour.

   e. Calculate the sum of 1000 numbers that are being input interactively using a loop.

   f. Display a message “You are an excellent student” for a student whose cumulative grade point average (CGPA) is 3.75 or above.

g. Calculate the circumference of a circle if the radius entered is a positive value.

h. Determine whether a number entered by a user is an even number.

i. MyHealth magazine is sold at RM5 each if at least 5 units of the magazine are purchased, and it is sold at the price of RM7 each otherwise. Calculate the price a customer has to pay after he enters the quantity of the magazines that he purchased.

j. Calculate water bill based on water consumption as given below:

Usage first 20 m3 - charge is 40 cents per m3
Additional 20 m3 - charge is 20 cents per m3
Extra usage - charge is 10 cents per m3

k. Program will determine the total and average price for a few items bought by a user. The number of items will be entered by user at the start of the program.

l. Calculate the average of a few students' height. The value -1 for height indicate the end of input.

4. Identify input, process, and output for the following problem statements:
   a. Calculator that allow user to perform multiplication and division of two numbers. Prompt user to input and display the output.

   b. Lily wants to calculate an average of a series positive numbers. The process will stop if the number entered has a negative value.

c. If the payment made my customer is over RM 15 000, display a message that the customer will be given a free air ticket to Bangkok. If the payment is between RM 10 000 to RM 15 000, display a message that the customer will be given a free air ticket to Pulau Langkawi. Otherwise, display a message that the customer will be given a free lunch at the Palace of Golden Horses Hotel.

d. Sheena lives in Ipoh. She has to commute from Ipoh to Gopeng 5 days a week and have to pay RM3.20 every time she goes through the toll. Calculate the toll amount to be paid by Sheena every month. (Assume 1 month = 4 weeks).

e. A Go Go Hypermart wants to organize an annual sale. The discount given is display in the table below.

Amount Paid (RM)	Discount
More than RM500	15%
RM200- RM500	10%
Below RM200	5%

Calculate and display the total discount and amount to be paid by customer after discount.

f. Prompt user to input in Kilobyte and convert the input into Megabyte, Gigabyte,Terabyte and Petabyte.

g. The price of a cup of coffee sold at MyKopitiam Café is based on the number of scoops of espresso coffee, sugar syrup and milk. The prices of espresso coffee, sugar syrup and milk are RM5, RM3 and RM0.20 per scoop respectively. Identify the input, process and output to calculate the price of a cup of coffee sold at MyKopitiam Café.

h. A contractor needs a program that calculates and displays the volume of a rectangular Swimming pool.

i. Calculate and display Ahmad’ age.

j. Mr Rahman plans to make reservation for a hotel room in Langkawi Island. The basic rate is RM250.00 per night and the sale and the service charge is 15% of the room rate. Determine the total payment if he books the hotel room for n nights. List the input, processes involved to calculate the total payment and the output.

k. Given a list daily temperatures for a certain number of days, design an algorithm to determine the maximum temperature. The first input is the number of days.

l. A program will accept a series of numbers from user and determine whether each number entered is an even or odd number. The program will stop when a user enters 0.